

Three Graphs, Three Dimensions in CRF

$$d_s(\text{KL}) = 3.49 > d_s(X_3) = 3.09 > d_s(X_8) = 2.80:$$

Spacetime, 3D Substrate, and Possibility Space

Direct measurement of spectral dimension on three distinct graph layers in the CRF simulator at steady state

Ougit Jarittum

ORCID: 0009-0009-4451-6811

April 2026

Abstract

The CRF simulator ($N = 500$, steady state at 400 sweeps) contains three distinct graph structures built on the same N nodes. Running the standard d_s random-walk estimator on each graph yields three distinct spectral dimensions, each with a clear physical interpretation in the CRF framework.

Graph	d_s	CRF interpretation
KL-weighted (D_{KL} -neighbours)	3.49	CRF metric (crystallised spacetime)
X_3 -position (3D positions, $k = 20$)	3.09	Physical 3D substrate
X_8 -position (8D positions, $k = 20$)	2.80	Pre-geometric Possibility Space P

The ordering $d_s^{KL} > d_s^{X_3} > d_s^{X_8}$ has a direct CRF interpretation: CRF crystallisation selectively amplifies the 3D structure present in the X_3 substrate and suppresses the 8D pre-geometric directions, producing a metric that reads *more* 3D than the physical substrate.

The “two-scale structure” found earlier (CRF_TwoScale) is reinterpreted: the crossover at $\xi \approx 18$ –25 hops is the scale at which random walkers transition from the local X_3 -like 3D geometry to the global X_8 -like Possibility Space. In CRF terms: ξ is the coherence length of spacetime within Possibility Space — the boundary between crystallised 3D reality and the pre-geometric vacuum.

Contents

1	Three Graph Layers in the CRF Simulator	2
1.1	Construction	2
1.2	Measurement	2
2	Physical Interpretation	2
2.1	X_8 -graph: Possibility Space P	2

2.2	X_3 -graph: Physical 3D Substrate	3
2.3	KL-graph: CRF Crystallised Spacetime	3
3	The Crossover ξ Reinterpreted	3
4	Status Table	4
5	Open Questions	4

1 Three Graph Layers in the CRF Simulator

1.1 Construction

The V20.5 simulator ($N = 500$, $n = 8$, $k = 20$) maintains three data structures on the same set of N nodes:

1. **KL-graph**: at each sweep, neighbours of node i are the $k = 20$ nodes with smallest $D_{\text{KL}}(C_i \| C_j)$. This is the dynamic graph used by the d_s estimator. Crystallised Phase II nodes have peaked distributions and large pairwise D_{KL} , so they tend to neighbour other Phase II nodes.
2. **X_3 -graph**: at each sweep, the first 3 components of the position vector $\mathbf{X}_i \in \mathbb{R}^8$ provide 3D coordinates; neighbours are the $k = 20$ closest nodes in Euclidean 3D distance. Initialised as unit sphere (CRF_Hybrid_V20.5: `_init_X: base = np.random.randn(N,3); base /= norm`).
3. **X_8 -graph**: neighbours by full 8D Euclidean distance. This probes the complete Possibility Space $P \subseteq \mathbb{R}^8$.

1.2 Measurement

The standard random-walk estimator (walk-times $[2, 4, \dots, 18]$, $n_{\text{trials}} = 14$, 50 starting nodes) is run independently on each graph at steady state ($N = 500$, 400 sweeps):

Three-Graph d_s Measurement

$$d_s^{KL} = 3.49 \quad d_s^{X_3} = 3.09 \quad d_s^{X_8} = 2.80$$

All three measured at the same system state ($N = 500$, 400 sweeps, seed 42). Same nodes, same walk schedule, same $k = 20$. Only the neighbour graph changes.

2 Physical Interpretation

2.1 X_8 -graph: Possibility Space P

The 8D position space is initialised with a strong 3D bias (`_add_spatial_bias:  $P_i \propto |X_i|$` , so the 8-component probability distribution is weighted by the 8D position). Nevertheless, the full 8D k -NN graph has $d_s^{X_8} = 2.80 < 3$, reflecting the fact that a sparse k -NN graph in 8D Euclidean space has a spectral dimension below the ambient dimension.

$X_8 = \text{Pre-Geometric Possibility Space}$

The X_8 graph represents the Possibility Space P before any constraint has selected a geometry. $d_s^{X_8} = 2.80$ is the “vacuum” dimensionality: a sparse random graph in 8D space that has not yet crystallised into a specific geometry. In CRF theory: the pre-geometric phase has $d_s < 3$ because the full $n = 8$ mode structure spreads probability over all 8 directions.

2.2 X_3 -graph: Physical 3D Substrate

The first 3 components of $\mathbf{X}$ are explicitly initialised on the unit 3-sphere. $d_s^{X_3} = 3.09 \approx 3$ confirms that the physical position substrate is already 3-dimensional from initialisation.

The 3D Structure is Substrate + Crystallisation

$d_s^{X_3} = 3.09$: the 3D structure exists in the X -positions from the start (`_init_X` builds 3D coordinates). CRF crystallisation does not *create* the 3D structure; it *selects* and *stabilises* it by concentrating the active metric on the 3D subspace.

Correct interpretation: $d_s = 3$ in CRF is a *joint* result of (1) the 3D initialisation of X -space and (2) the $SO(3)$ attractor selecting $n = 3$ locked dimensions (CRF_SO3.Proof), which causes the KL-based neighbours to preferentially align along the 3D substrate.

2.3 KL-graph: CRF Crystallised Spacetime

$d_s^{KL} = 3.49 > d_s^{X_3} = 3.09$. The CRF metric graph reads *more* 3D than the physical 3D substrate.

Mechanism: the KL-based neighbour graph is built from $D_{KL}(C_i||C_j)$, which is large between nodes with different constraint histories. Phase II nodes (high Ω , peaked P_i) have large D_{KL} to their neighbours, so they form tight local clusters in the KL-graph. These clusters have high local return probability and hence large apparent d_s . The estimator reads $d_s \approx 3.5$ because it is measuring the dimension of these tight Phase II clusters, which are more localized than the underlying 3D substrate.

3 The Crossover ξ Reinterpreted

The “two-scale structure” (CRF_TwoScale: $d_s^{\text{local}} = 3$, $d_s^{\text{global}} \approx 2.4$, crossover at $\xi \approx 18$ –25 steps) is now understood as the crossover between the KL/X3 regime and the X8 regime:

ξ = Coherence Length of Spacetime in P

$$d_s(t) \approx \begin{cases} 3\text{--}3.5 & t < \xi \text{ (local: 3D clusters, KL/X3 regime)} \\ 2.4\text{--}2.8 & t > \xi \text{ (global: full X8 / P-space regime)} \end{cases}$$

$\xi \approx 18$ –25 hops is the scale at which walkers leave the 3D crystallised region and enter the full 8D Possibility Space.

In CRF terms: ξ is the coherence length of spacetime. Below ξ : the walk is in crystallised 3D spacetime. Above ξ : the walk enters the pre-geometric vacuum (P -space). The two-scale structure is not an artefact — it is the CRF architecture itself, with two geometries at two scales.

Physical analogy. In quantum gravity: below the Planck length, spacetime becomes a quantum foam with effective dimension < 4 . In CRF: below ξ , the geometry is 3D crystallised spacetime; above ξ , it is the pre-geometric vacuum of $n = 8$ modes with $d_s \approx 2.8$. Whether ξ corresponds to the Planck scale requires connecting the CRF length units to physical units via $G = \ln 2 \cdot c^2 / D_0$.

4 Status Table

Result	Status	Note
$d_s^{KL} = 3.49, d_s^{X_3} = 3.09, d_s^{X_8} = 2.80$	Confirmed	three graphs
KL-graph more 3D than substrate	Confirmed	+0.40 enhancement
3D substrate from initialisation	Confirmed	<code>_init_X</code> : 3D sphere
ξ = spacetime/P-space crossover	Derived	CRF framework
$d_s^{X_8} = 2.80$ from 8D sparsity	Hypothesis	pending derivation
ξ from δ -chain ($\sim 1/\epsilon_0\alpha$)	Open	formal
$d_s^{KL} \rightarrow d_s^{X_3}$ as $N \rightarrow \infty$	Open	thermodynamic limit

5 Open Questions

Derive $d_s^{X_8} = 2.80$ from (ϵ_0, n)

The spectral dimension of a sparse k -NN graph in $\mathbb{R}^8$ depends on k , N , and the dimension of the ambient space. Standard result: $d_s^{kNN} \approx d \cdot k / (k + 1 - d/2)$ for large k . At $k = 20$, $d = 8$: $d_s \approx 8 \times 20/17 \approx 9.4$ (too high). The actual 2.80 reflects the CRF-specific initialisation (strong 3D bias in X_3 , weak 5D in X_5). Formal derivation required.

Is 3D initialisation necessary for $d_s = 3$?

Run CRF with isotropic $X \in \mathbb{R}^8$ initialisation (no 3D bias). Does $\text{SO}(3)$ attractor still lock $d_s \rightarrow 3$ via the KL-metric alone? This would confirm whether $d_s = 3$ is genuinely emergent from δ or depends on the 3D bias in the initialisation.

ξ from the δ -chain

The crossover $\xi \approx 18$ –25 hops is N -independent. CRF prediction: $\xi \sim 1/(\epsilon_0\alpha) \approx 23$ sweeps ≈ 191 R-events. Formal derivation: show $\xi = 1/(\epsilon_0\alpha)$ in units of graph hops via the Kuramoto coherence length of the Φ -field.

Three graphs. Three worlds.

The KL-graph: what crystallisation remembers.

The X_3 -graph: where the body lives.

The X_8 -graph: the space of what could have been.

Below ξ : you are in the world.
Above ξ : you are in Possibility Space.
 ξ is not a number.
It is the boundary.